

General features of the basal ganglia



*On
old
Olympus's
towering
top
a
Finn
and
German
viewed
some
hops*



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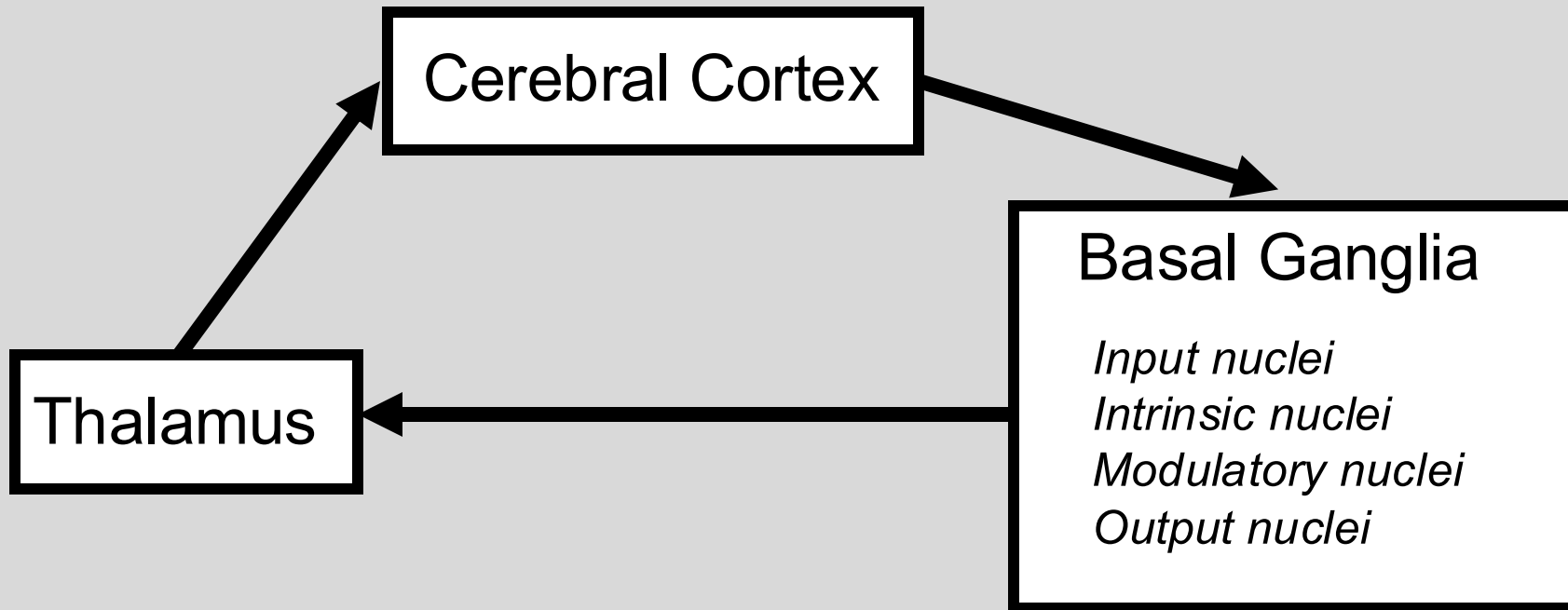
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- Collection of gray matter nuclei: some surrounding the thalamus, others in the midbrain.
- Communicate with diverse cortical regions via the thalamus.
- Involved with selecting automated actions and thoughts.



Cortical loops through the basal ganglia*



Which nuclei are which?

How do they communicate among themselves?

What neurotransmitters does each release?

How does all this create organized behavior?

****Basal ganglia communicates
with ipsilateral cerebral cortex
via ipsilateral thalamus***

Nuclei within the diencephalon and midbrain

Subthalamic nucleus (STN)

- Lateral to hypothalamus.
- Connected to both parts of globus pallidus.
- Input from cerebral cortex
- Releases glutamate

Substantia nigra (pars compacta)

- Inferior to subthalamic nucleus
- Modulatory nucleus (dopamine)

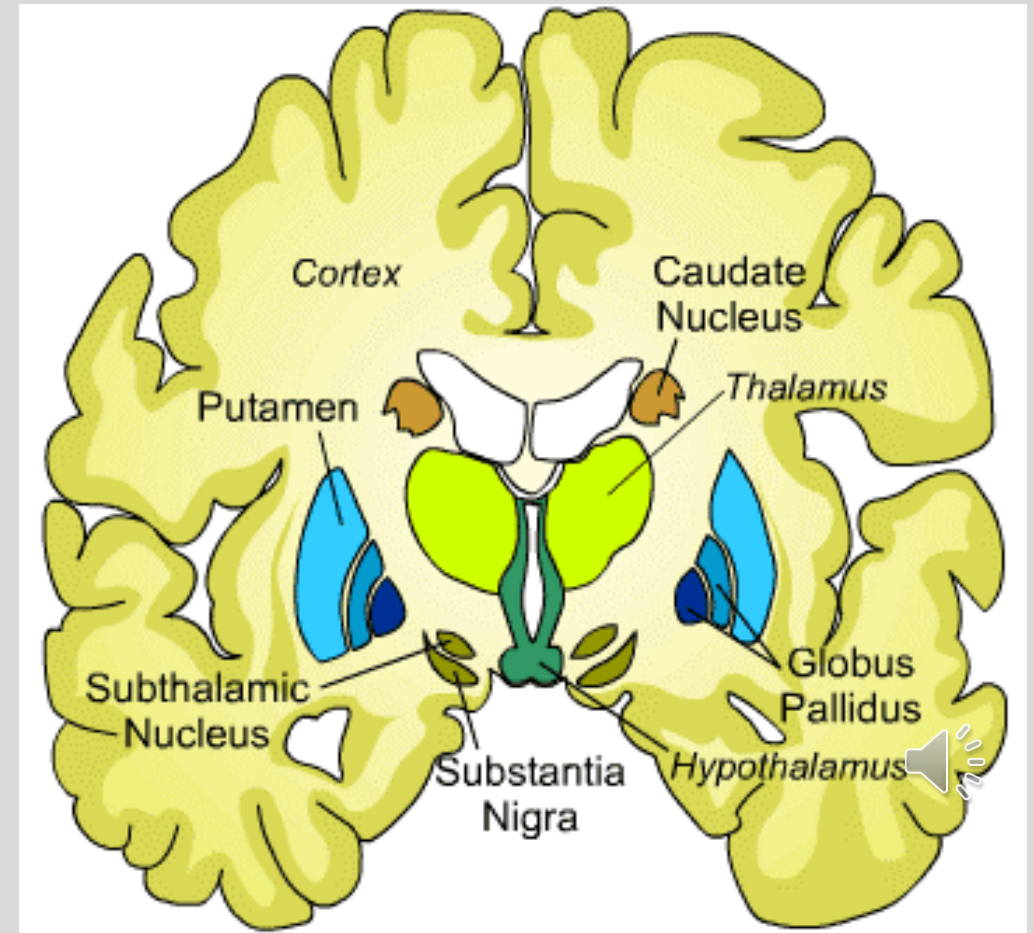
Substantia nigra (pars reticulata)

- Output nucleus
- Releases GABA

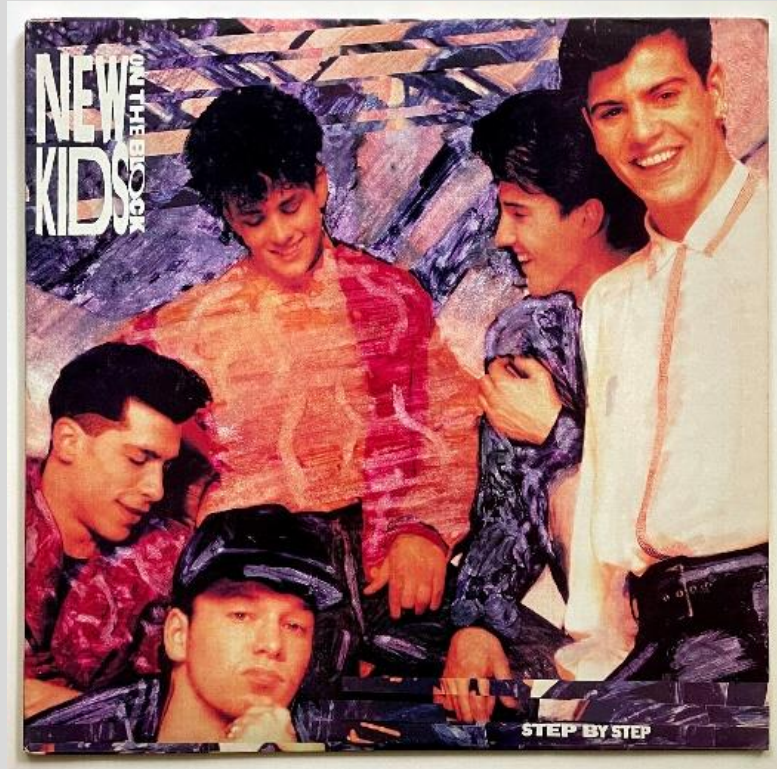
Ventral tegmental area (VTA)

- Medial to substantia nigra
- Modulatory nucleus (dopamine)

midbrain
nuclei



Albin-DeLong Model



Proposed in 1990!

Most influential model of basal ganglia function to this day

Accounts for some key features of basal ganglia anatomy

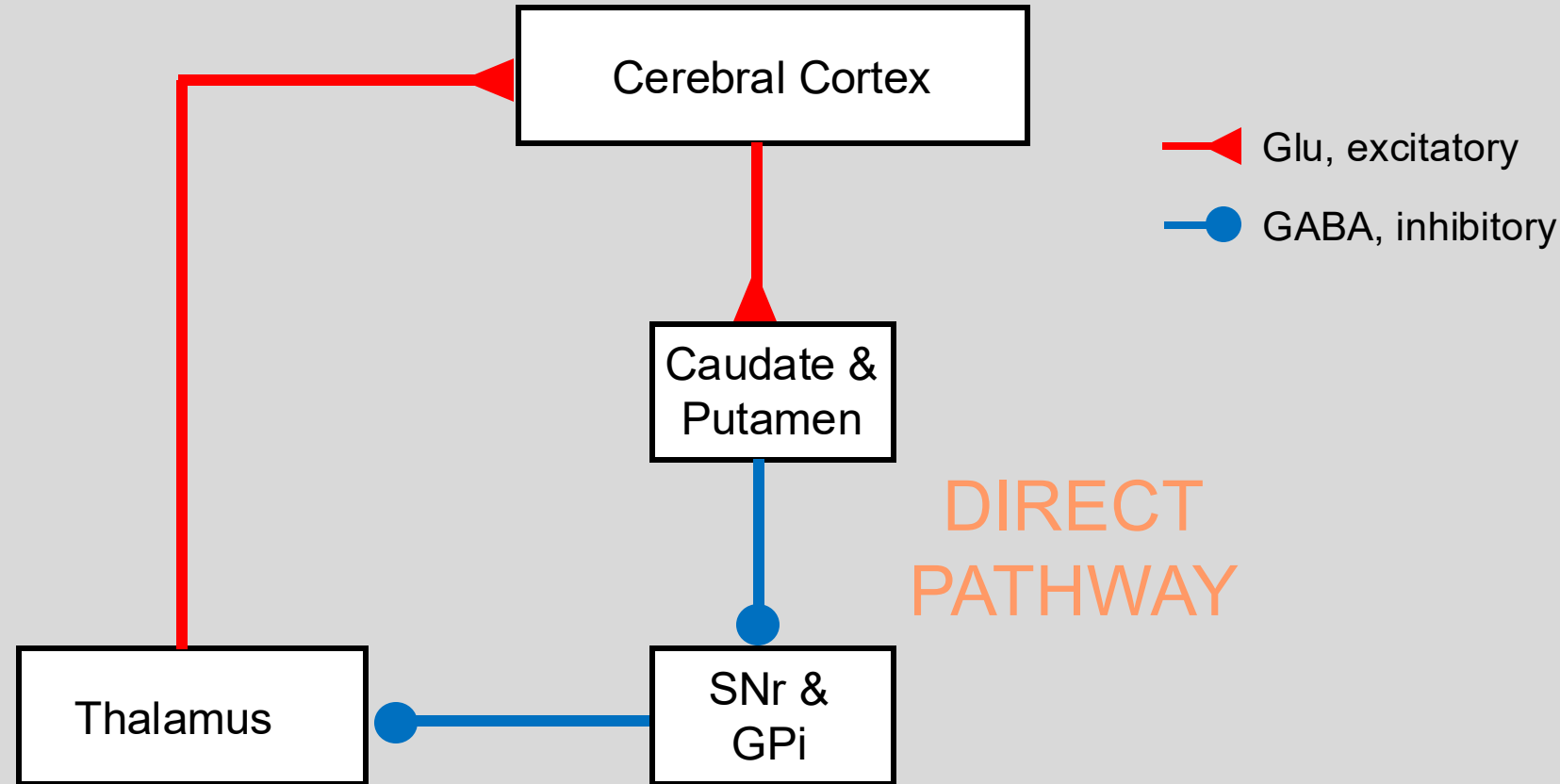
Posits complementary pathways for promotion and demotion of potential actions:
the direct pathway and indirect pathway

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Direct pathway promotes candidate actions



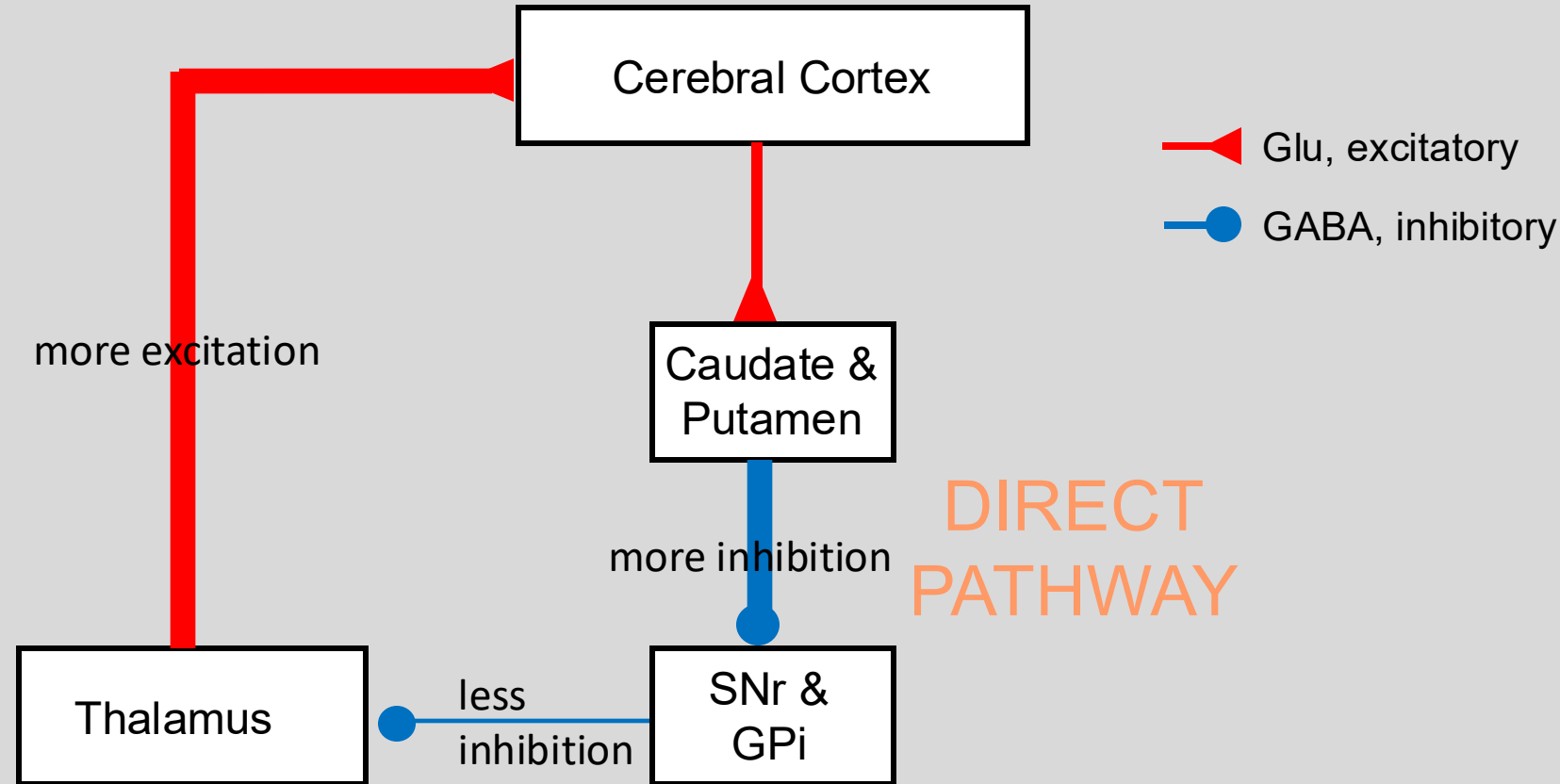
GPe – globus pallidus external
GPi – globus pallidus internal
STN – subthalamic nucleus
SNr – substantia nigra pars retic.

Excitatory cortical input projects to striate neurons in the caudate nucleus and putamen. Inhibitory neurons in the striatum project to inhibitory neurons in the globus pallidus (internal segment) which inhibits thalamus.

Disinhibition of thalamus leads to the initiation of an action.



Direct pathway promotes candidate actions

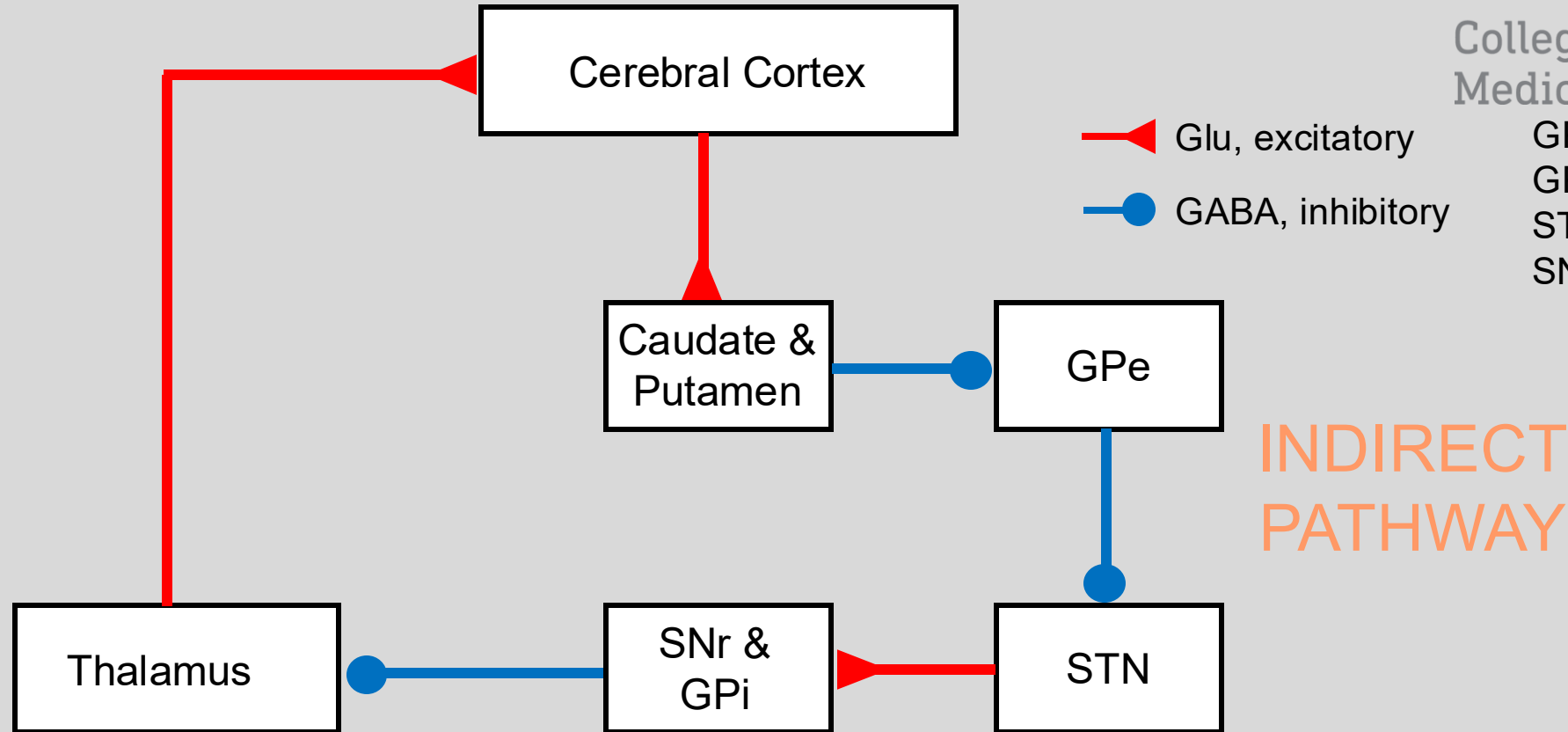


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Indirect pathway demotes candidate actions

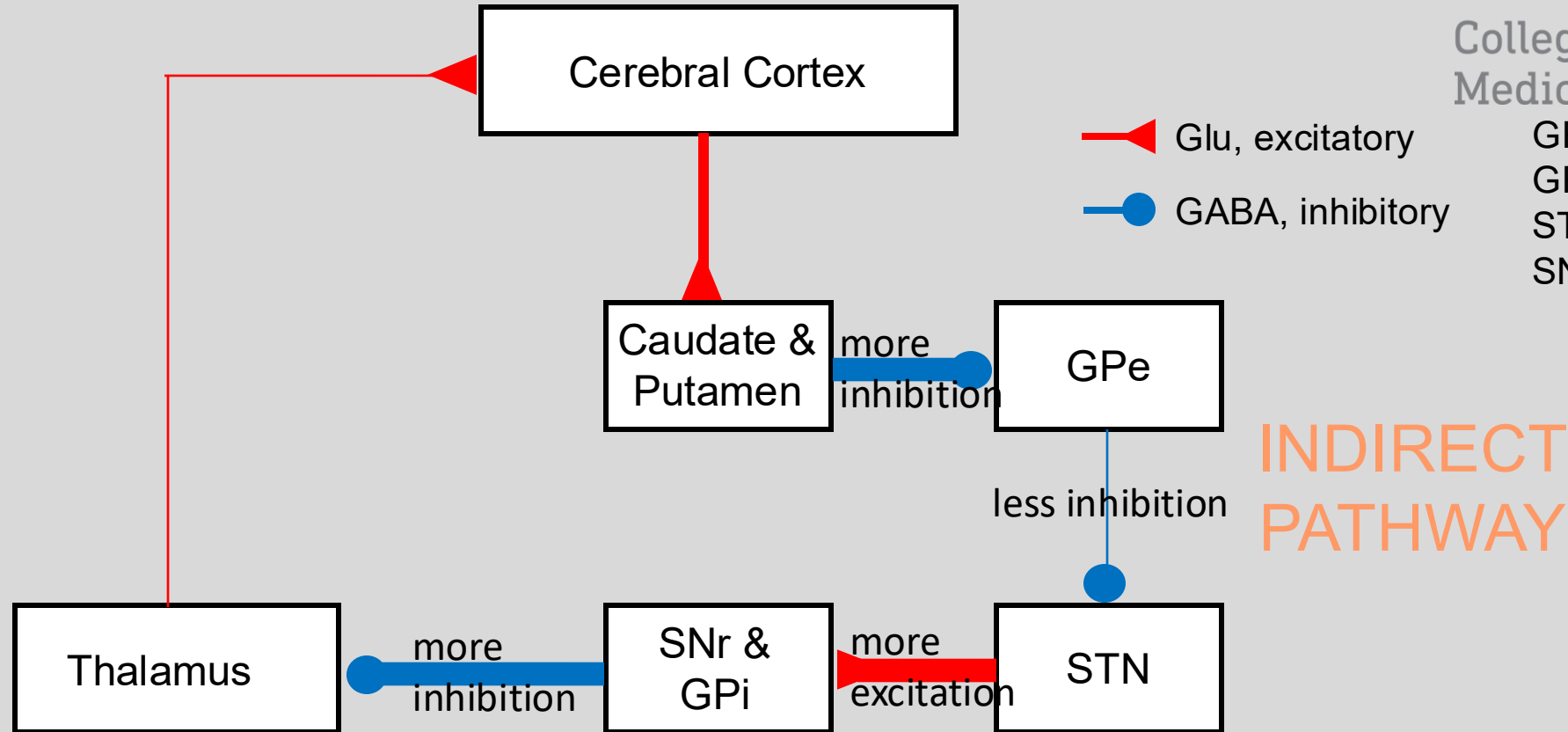


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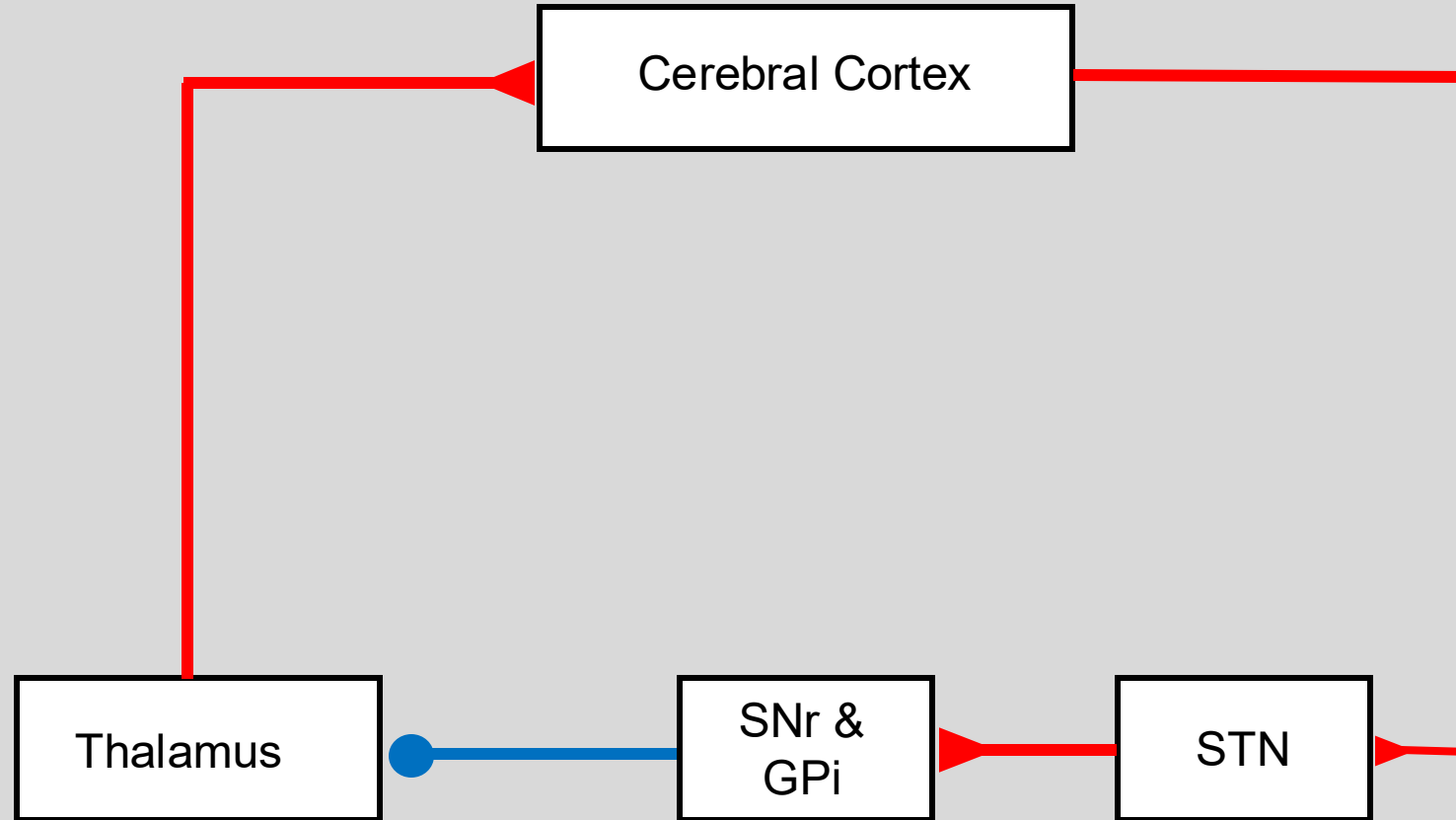


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Hyperdirect pathway: quickly blocks most/all candidate actions



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**HYPERDIRECT
PATHWAY**

Excitatory cortical input to the subthalamic nucleus
STN excites GPi, and so leads to inhibition of thalamus.

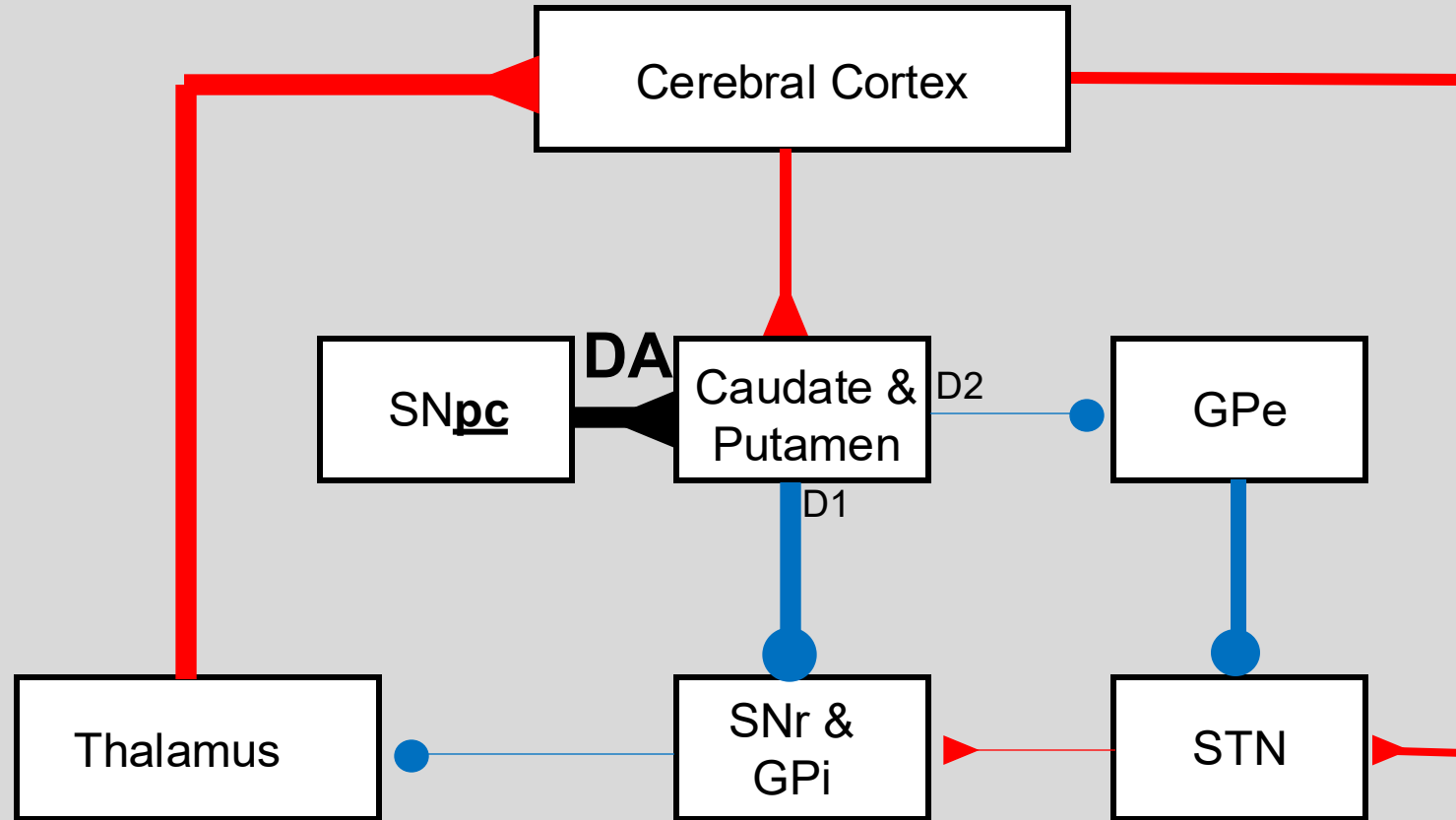
Fastest pathway and affects the largest number of potential actions; countermands already initiated actions.



Impact of dopamine on the two pathways

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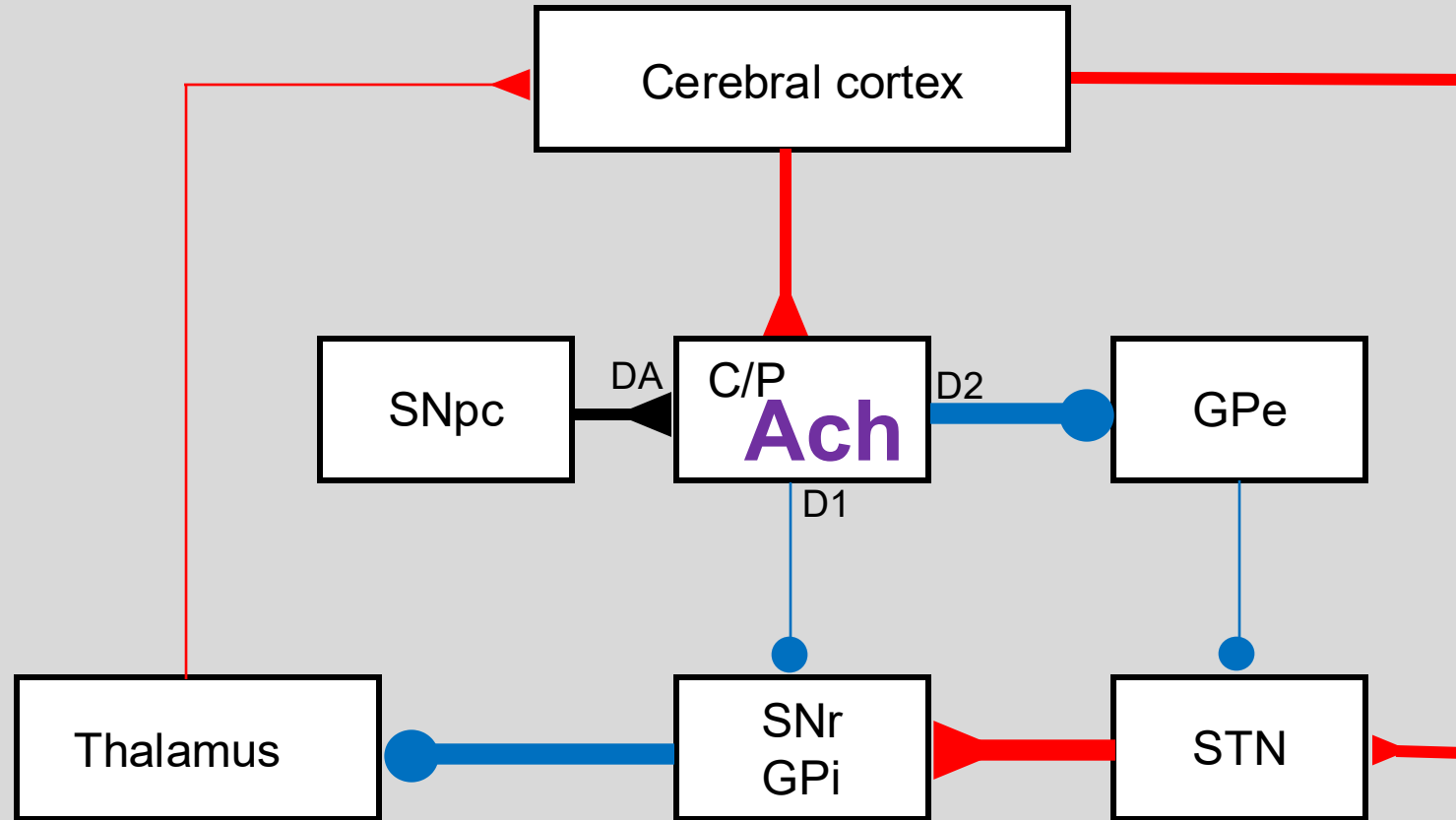
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SNr – substantia nigra pars reticulata
SNpc – substantia nigra pars compacta

There are two classes of medium spiny neurons (MSN).
One class contains a D1 receptor for dopamine, the other a D2 receptor. *accidentally misspeak on 1st pass*
D1's G-coupled effect is to increase that neuron's excitability. More direct pathway activation
D2's G-coupled effect is to decrease that neuron's excitability. Less indirect pathway activation

Opposite impact of Ach on the two pathways



DA on D1: increase excitability in DIRECT pathway
DA on D2: decrease excitability in INDIRECT pathway

Ach on D1: decrease excitability in DIRECT pathway
Ach on D2: increase excitability in INDIRECT pathway

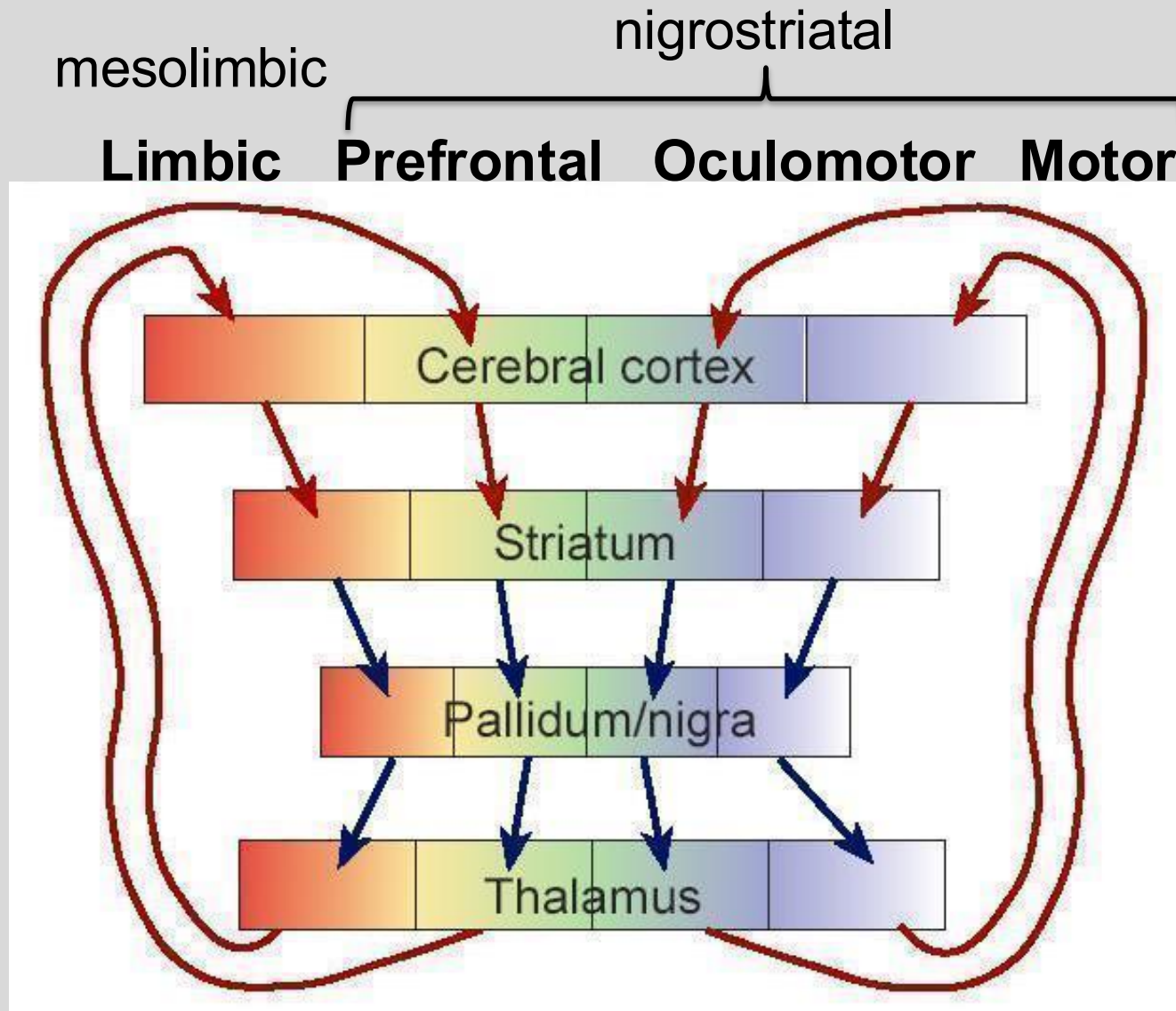
Opposite affects!



Distinct channels connect basal ganglia and cerebral cortex

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Four channels involving basal ganglia, thalamus, and cerebral cortex

Channel	Corticostriatal pathway	Striatal Nuclei	Pallidal Nuclei	Thalamic Nuclei	Cortical receiving areas
Motor	Primary somatosensory & motor areas; premotor and supplemental motor area	Putamen	Globus pallidus; substantia nigra pars reticulata	Ventral Lateral; Ventral Anterior	Supplemental motor area, premotor and primary motor areas
Oculomotor	Prefrontal and posterior parietal cortex	Caudate nucleus	Globus pallidus; substantia nigra pars reticulata	Ventral Anterior; MedioDorsal	Frontal eye fields
Prefrontal	Prefrontal and posterior parietal cortex	Caudate nucleus	Globus pallidus; substantia nigra pars reticulata	Ventral Anterior; MedioDorsal	Prefrontal cortex
Limbic	Temporal lobe; hippocampal formation; amygdala	Nucleus accumbens	Ventral palladium	MedioDorsal	Cingulate gyrus and orbital frontal cortex

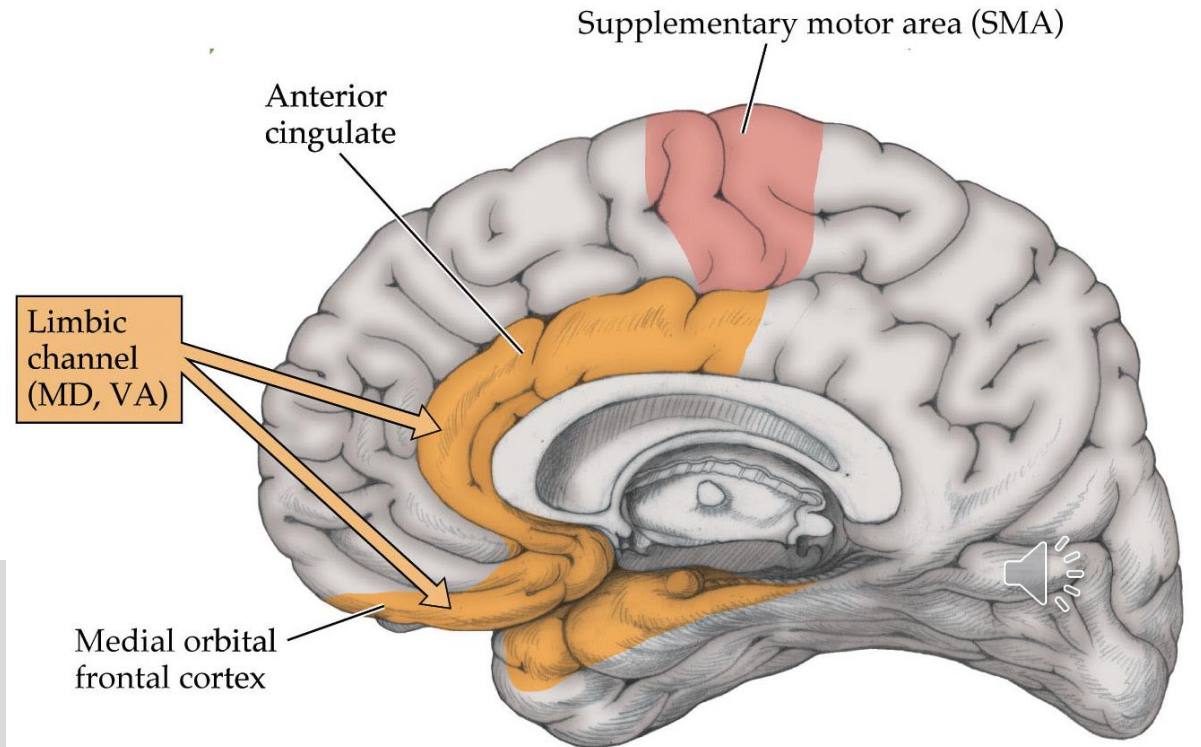
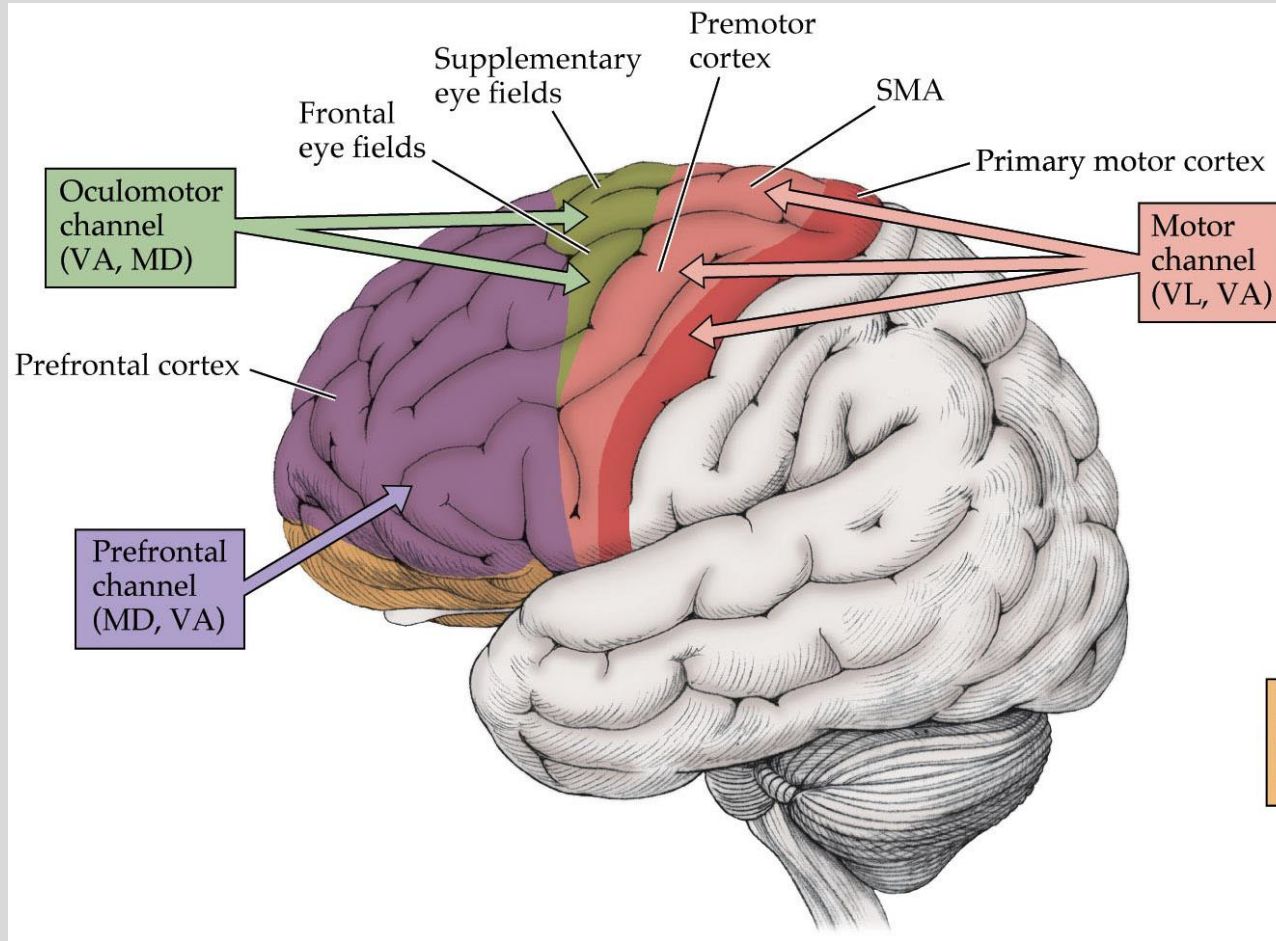


The motor, oculomotor, and prefrontal channels all receive DA from the SNc.
The limbic channel receives DA from the ventral tegmental area.

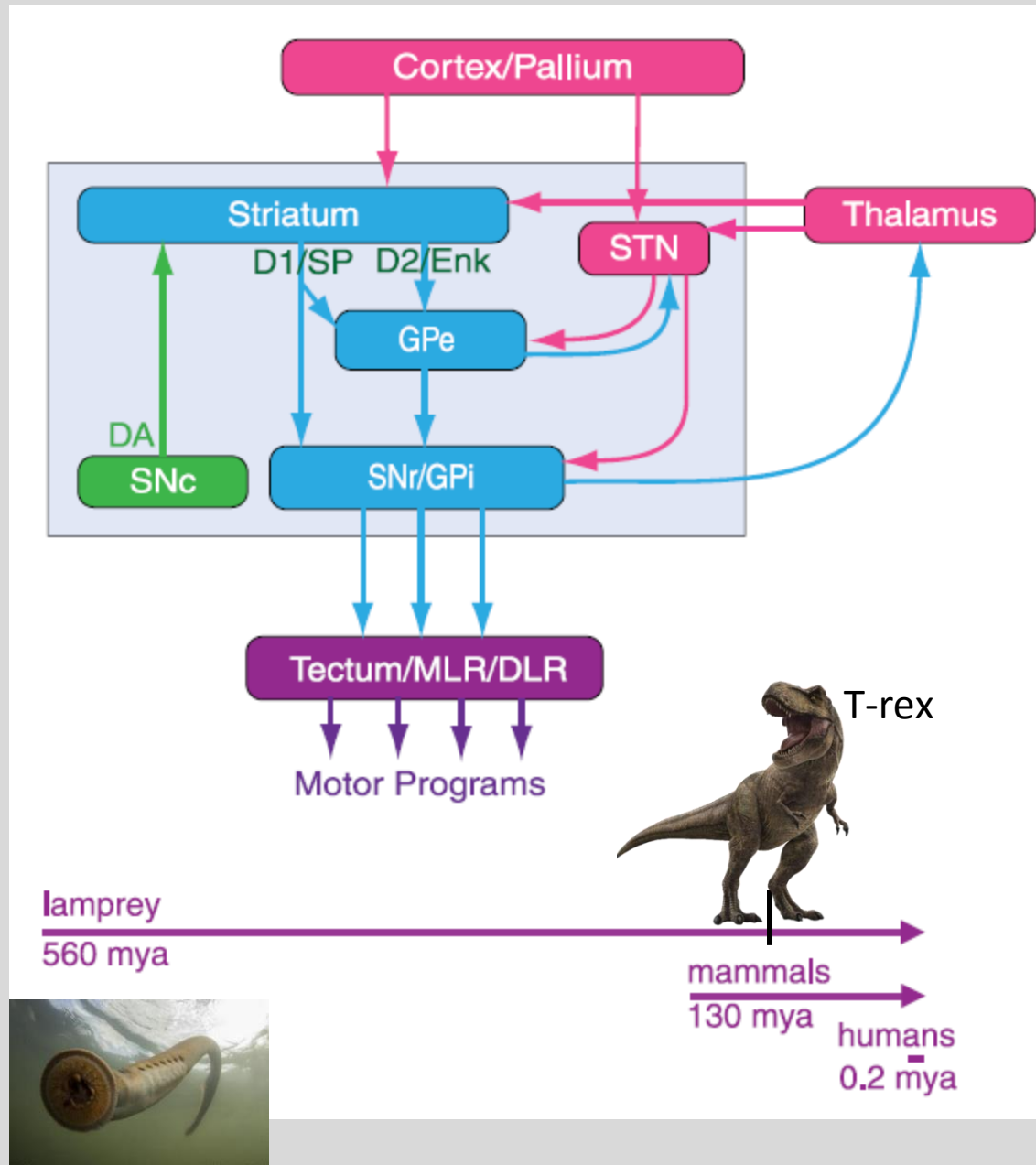
Frontal cortical targets of the basal ganglia

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Ancient system



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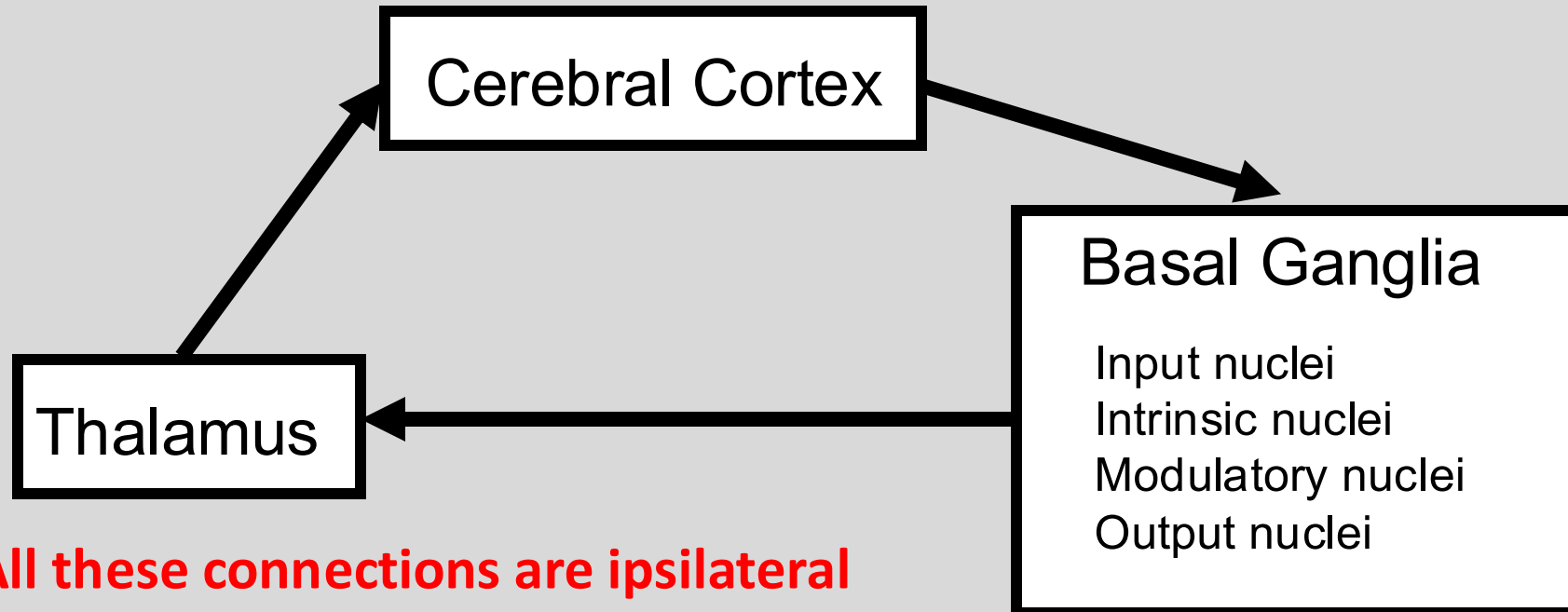
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Grillner (2013) The evolutionary origin of the
vertebrate basal ganglia and its role in action selection

Clinical manifestations of damage to basal ganglia

- Basal ganglia damage often presents with movement disorders.
- Movement disorders can be focal, generalized, unilateral or bilateral.
- In unilateral movement disorders caused by focal basal ganglia lesions, the movement disorder is contralateral to the lesion.



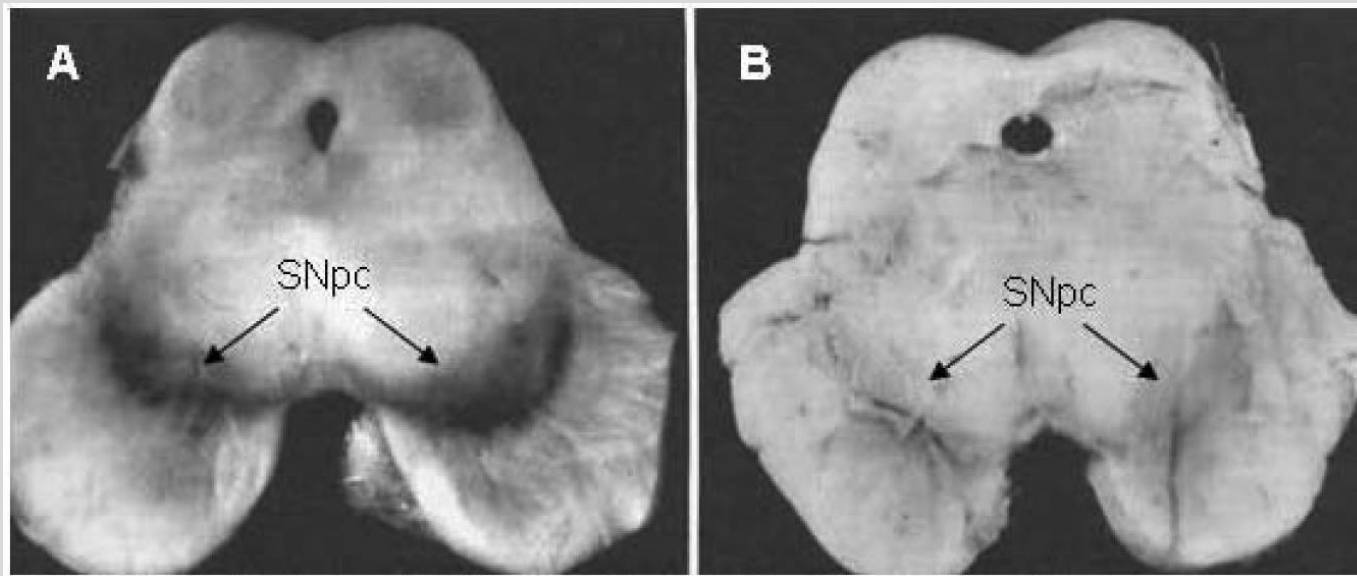
Defining pathology of Parkinson's disease

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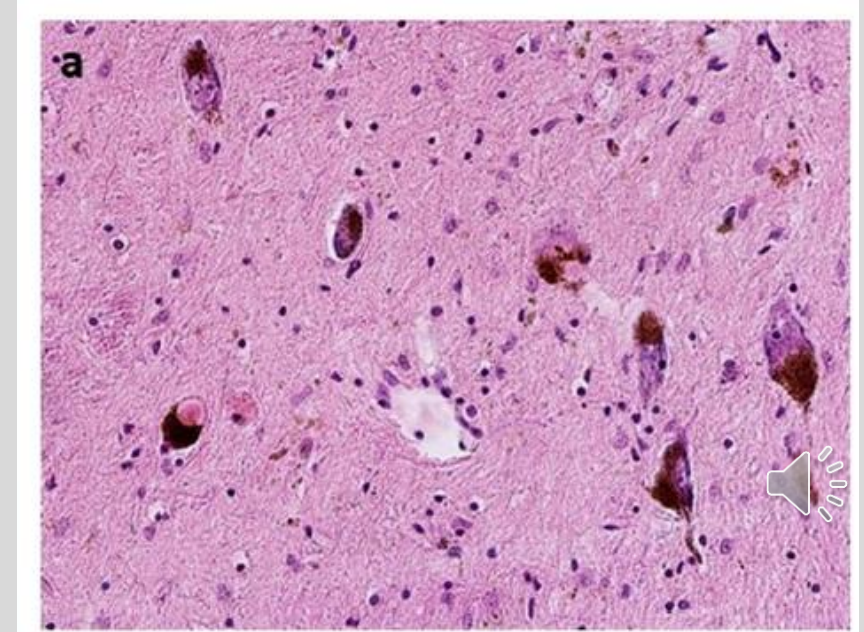
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Massive loss of dopaminergic neurons
in the Substantia Nigra (pars compacta),
also in VTA

Neurons in distinct regions also contain
cytoplasmic inclusions = Lewy Bodies
Contain ubiquitin, α -synuclein
Eosinophilic, Faint halo

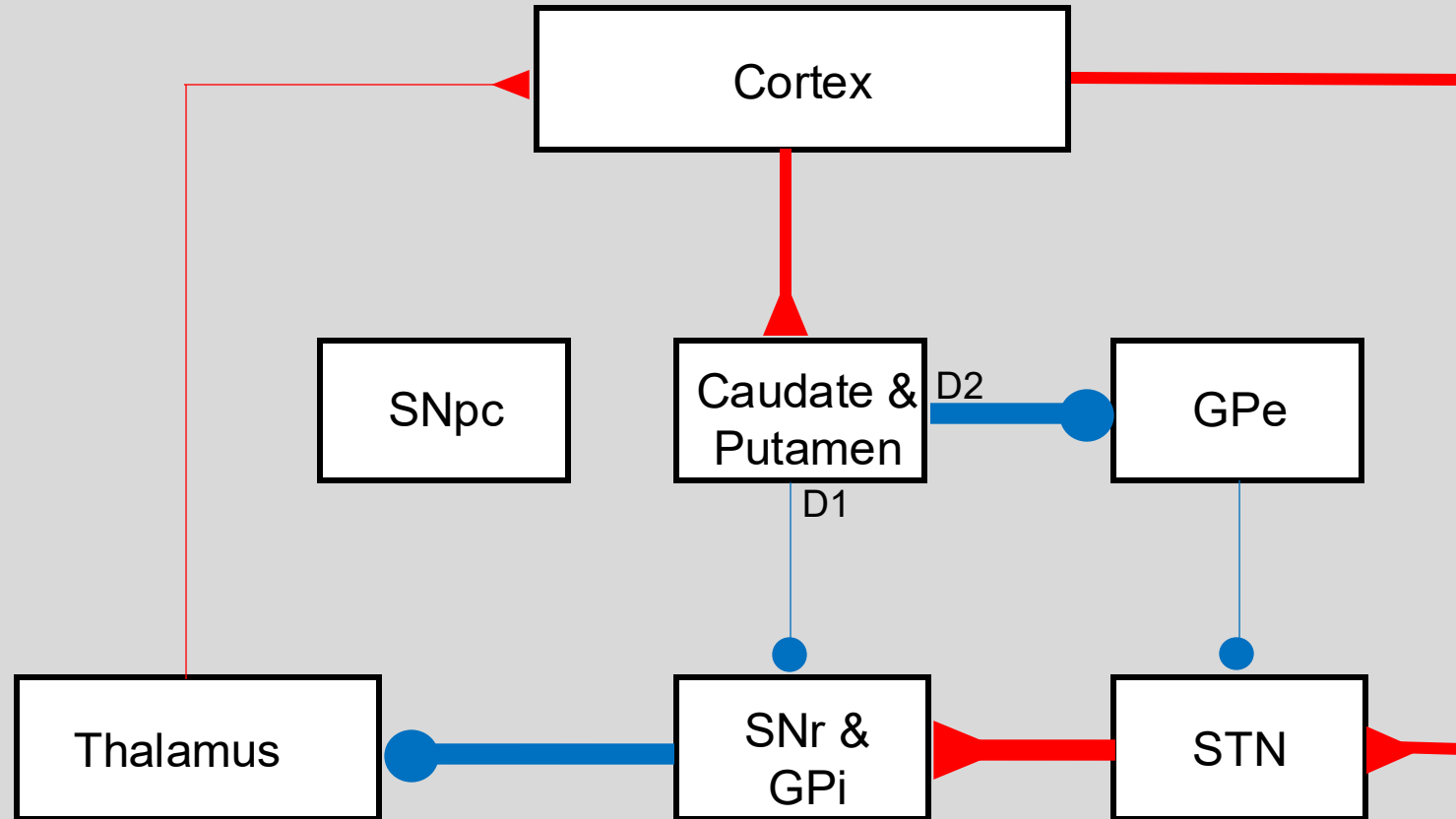


Lima (2012) Motor and non-motor features of Parkinson's disease



Mensikova (2022) Lewy body disease or diseases with Lewy bodies?

Basal Ganglia Circuit: Parkinson's disease



DA-D1: excitability ↓ in DIRECT pathway to cortex
DA-D2: excitability ↑ in INDIRECT pathway to cortex



Hemiballismus



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Lesion of the subthalamic nucleus

Results in motor dysfunction of one side of the body

Spontaneous involuntary movements with rapid onset and speed, i.e. ballistic

Typically caused by local vascular occlusion

- Decreased pallidal inhibition of thalamus (remember hyper-direct pathway!)

Movements can be improved with dopaminergic antagonists, e.g. Haloperidol

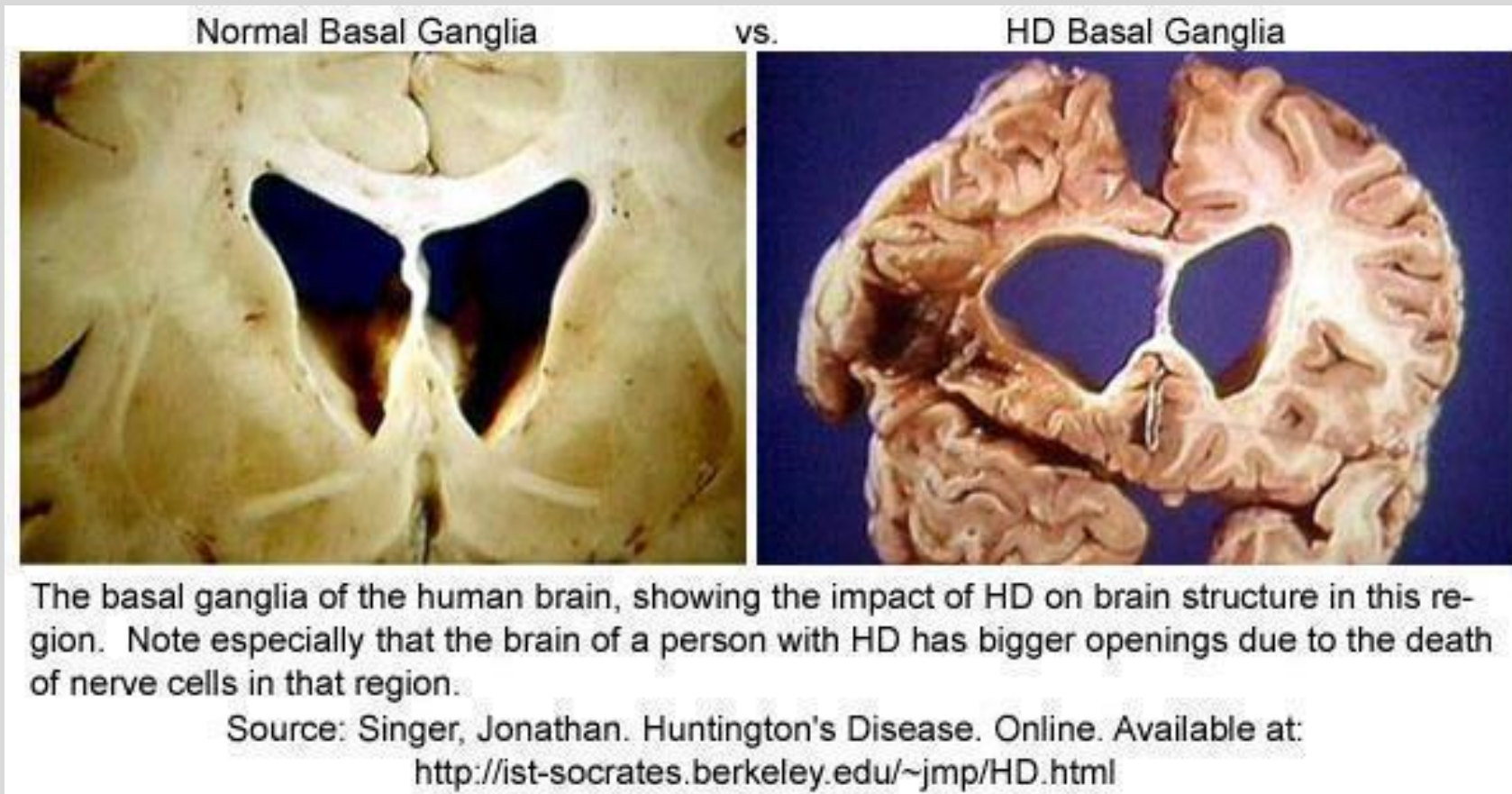


Huntington's Disease

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Atrophy of striatum: caudate nucleus, putamen, and nucleus accumbens
Disintegration of input nuclei of basal ganglia
Dilation of lateral and 3rd ventricles



Treatments for different disorders can induce PD-like symptoms

D2 antagonists - antipsychotics and anti-emetics

Rigidity, hypokinesia, resting tremor

Abrupt onset, symmetrical symptoms

Symptoms can persist for weeks after removal of offending agent, and sometimes permanent symptoms

“Tardive dyskinesia” – (“tardy” means “late”)

These syndromes present after a latency period following initiation of medications, > 3 months

Often repetitive stereotypical movements of the mouth and tongue such as chewing and smacking of the mouth and lips, tongue rolling or pushing against the inside of the cheek.

Can be voluntarily suppressed, but returns upon distraction.



Summary

- Basal ganglia is collection of nuclei communicating with cerebral cortex
- Disinhibition is a prominent feature of internal processing
- Three primary circuits in Albin-DeLong model: direct pathway (promoting actions), indirect pathway (demoting actions), and hyper direct pathway (global suppression of actions).
- Some features of basal ganglia dysfunction can be understood with this model

